

Supplementary Materials

Prodromal Critical Cascade in Parkinson’s Disease: A SymC Framework for Early Detection,
Phenotype Dynamics, and Precision Intervention

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Overview

The Prodromal Critical Cascade (PCC) is a general five-stage failure ontology derived from the Symmetrical Convergence (SymC) framework. It describes how adaptive systems approach critical failure through a predictable sequence of dynamical transitions, each characterized by distinct signatures in the damping ratio $\chi(t) = \gamma(t)/(2|\omega(t)|)$.

The PCC framework is domain-agnostic: the same cascade structure applies to neurodegeneration, seismic rupture, tumor escape, viral rebound, and engineered system failure. Domain-specific applications differ only in the physical interpretation of γ (damping/dissipation) and ω (characteristic frequency/drive), not in the mathematical structure of the cascade itself.

1 S1: The Universal Stability Equation

1.1 S1.1 The Core Principle

All complex adaptive systems are governed by a single dimensionless ratio:

$$\chi(t) = \frac{\mathcal{D}(t)}{2 \cdot \mathcal{E}(t)} \tag{1}$$

where:

- χ = **Adaptive Susceptibility Ratio** (distance from chaos or rigidity)
- $\mathcal{D}$ = **Dissipation** (braking force: inhibition, viscosity, damping, friction)
- $\mathcal{E}$ = **Energy/Drive** (driving force: frequency, metabolic rate, stress, dopamine)

The adaptive window $0.8 \leq \chi \leq 1.0$ represents the regime where systems maximize information transfer while maintaining structural integrity. Disease, failure, and instability arise when systems drift outside this window:

- $\chi < 0.8$ (Underdamped/Chaotic): System oscillates destructively, vibrates toward entropy
- $\chi > 1.2$ (Overdamped/Rigid): System stops producing information, freezes into stasis

1.2 S1.2 Domain Mapping

The universal equation takes different “costumes” across domains while the physics remains identical. Table 1 provides the cross-domain mapping.

Table 1: The SymC Rosetta Stone: Universal Variables Across Domains

Domain	Driver ($\mathcal{E}$)	Damper ($\mathcal{D}$)	$\chi < 0.8$	$\chi > 1.2$
Parkinson’s	Striatal drive (ω)	BG inhibition (γ)	Tremor	Rigidity/Freezing
Alzheimer’s	Neural firing (ω)	Tissue stiffness (G')	MCI/Anxiety	Dementia/Apathy
Addiction	Dopamine surge (ω_{DA})	PFC inhibition (γ_{PFC})	Binge/Craving	Withdrawal/Anhedonia
Geophysics	Tectonic stress (σ)	Viscosity/Friction (μ)	Tremor/Eruption	Locked fault
Computation	Signal (S)	Noise/Entropy (N)	Overfitting	Underfitting

1.3 S1.3 The Paradigm Shift

The SymC framework redefines disease:

Disease is not a foreign invader; it is simply the system drifting into the underdamped zone ($\chi < 0.8$) or the overdamped zone ($\chi > 1.2$).

This reframes every therapeutic goal from “fight the disease” to “restore the adaptive window.” L-DOPA, DBS, addiction treatment, and tectonic stress management all function as χ -modulators: interventions that steer systems back toward $\chi \approx 0.9$.

FigS2_Universal_Phase_Space.pdf

Figure 1: **The Universal Stability Manifold.** Phase space with drive ($\mathcal{E}$) vs. damping ($\mathcal{D}$). Diagonals = constant χ . Green band ($0.8 \leq \chi \leq 1.0$): adaptive window. Blue (overdamped): freezing, locked faults. Red (underdamped): tremor, rupture.

2 S2: The Five-Stage PCC Ontology

2.1 S2.1 Stage Definitions

The Prodromal Critical Cascade proceeds through five sequential stages, each representing a distinct failure mode in the system's control architecture:

1. **Stage 1: Gain Failure**

Definition: Attenuation of sensory or input channels. The system's ability to detect perturbations degrades before any controller malfunction occurs.

χ signature: χ remains within the adaptive window ($0.8 \leq \chi \leq 1.0$), but signal-to-noise ratio decreases. The system compensates successfully but with reduced margin.

Detectability: Subclinical. Measurable only with sensitive instrumentation or longitudinal tracking.

2. Stage 2: Damping Failure

Definition: Loss of inhibitory or dissipative control. The feedback mechanisms that stabilize the system begin to degrade.

χ signature: χ variability increases. The system fluctuates between underdamped ($\chi < 0.8$) and adaptive states. Intermittent oscillatory episodes appear.

Detectability: Subclinical but detectable. Variance-based biomarkers become abnormal before mean values shift.

3. Stage 3: Material Fatigue

Definition: Accumulation of structural damage or pathological substrate. The physical or biological medium sustaining the control system degrades.

χ signature: χ begins systematic drift from the adaptive window. The system can no longer fully compensate for perturbations.

Detectability: Biological markers or physical measurements confirm substrate degradation. This stage typically marks the transition from reversible to partially irreversible damage.

4. Stage 4: Resonance Nucleation

Definition: Emergence of pathological oscillations or clustering phenomena. The system enters a regime where small perturbations trigger amplified responses.

χ signature: χ falls below 0.8 (underdamped instability) or rises above 1.2 (overdamped rigidity), depending on the failure mode. Oscillatory bursting or critical slowing becomes measurable.

Detectability: Clinical or operational detection threshold. Symptoms or failure precursors become apparent.

5. Stage 5: Resonance Lock

Definition: Irreversible capture into a pathological attractor. The system cannot return to the adaptive window without external intervention.

χ signature: χ diverges ($\chi \rightarrow 0$ or $\chi \rightarrow \infty$ depending on failure direction). The system loses the capacity for adaptive response.

Detectability: Clinical diagnosis or system failure. The cascade is complete.

FigS1_Universal_PCC.pdf

Figure 2: **The Universal Critical Cascade (PCC).** Time-series visualization of the five-stage prodromal cascade showing χ trajectory from stable operation through failure. **Stage 1 (Gain Failure):** χ remains stable but noise increases. **Stage 2 (Damping Failure):** Variance spike marks the optimal intervention window. **Stage 3 (Material Fatigue):** Systematic drift begins; preclinical divergence is detectable. **Stage 4 (Resonance Nucleation):** χ exits the adaptive window with high-amplitude oscillations. **Stage 5 (Resonance Lock):** System captured in pathological attractor. Bottom rows show domain-specific manifestations across Physics (friction/viscosity), Parkinson’s (hyposmia to motor lock), Oncology (immune drop to resistance), and Addiction (tolerance to relapse).

2.2 S2.2 The Intervention Window

The PCC framework identifies an **Optimal Intervention Window** between Stages 2 and 3, where:

- Damping failure is detectable but structural damage is minimal

- Interventions can restore χ to the adaptive window
- The cascade remains reversible

After Stage 3 (Material Fatigue), interventions become progressively less effective as irreversible damage accumulates. The goal of PCC-informed monitoring is detection of Preclinical Divergence within the intervention window.

3 S3: Mathematical Structure

3.1 S3.1 The Central Equation

The PCC is governed by the damping ratio:

$$\chi(t) = \frac{\gamma(t)}{2|\omega(t)|} \quad (2)$$

where:

- $\gamma(t)$ = instantaneous decay rate (damping, dissipation, friction)
- $\omega(t)$ = instantaneous characteristic frequency (drive, oscillation, stiffness-derived)

The adaptive window is defined as $0.8 \leq \chi \leq 1.0$, with $\chi = 1$ marking the critical-damping boundary.

3.2 S3.2 Stage-Specific χ Dynamics

Table 2: χ trajectory through PCC stages

Stage	Mean χ	χ Variance	Trajectory
1. Gain Failure	0.85–0.95	Low	Stable within window
2. Damping Failure	0.80–1.0	Elevated	Intermittent excursions
3. Material Fatigue	Drifting	Elevated	Systematic departure
4. Resonance Nucleation	< 0.8 or > 1.2	High	Oscillatory/rigid
5. Resonance Lock	$\rightarrow 0$ or $\rightarrow \infty$	Divergent	Attractor capture

Key insight: Variance increases *before* mean χ shifts. This is the critical slowing signature that enables early detection.

3.3 S3.3 Estimation Uncertainty and Confidence Bounds

Reliable χ extraction from empirical signals requires quantification of estimation uncertainty. The error in χ estimation depends on both signal quality and epoch duration.

3.3.1 Error Propagation Analysis

Given $\chi = \gamma/(2|\omega|)$, the relative error propagates as:

$$\frac{\delta\chi}{\chi} = \sqrt{\left(\frac{\delta\gamma}{\gamma}\right)^2 + \left(\frac{\delta\omega}{\omega}\right)^2} \quad (3)$$

For Hilbert-based estimation, the dominant error source is the derivative of log-amplitude (for γ) and phase unwrapping artifacts (for ω).

3.3.2 Modality-Specific Uncertainty Bounds

Based on theoretical analysis and Monte Carlo validation with synthetic signals:

LFP beta (13–30 Hz):

- SNR ≥ 5 , epoch ≥ 60 s: $\text{RMSE}(\chi) \approx 0.05\text{--}0.07$
- SNR ≥ 10 , epoch ≥ 120 s: $\text{RMSE}(\chi) \approx 0.03\text{--}0.05$
- Primary error source: Low-frequency drift in amplitude envelope

Tremor accelerometry (3–8 Hz):

- SNR ≥ 5 , epoch ≥ 30 s: $\text{RMSE}(\chi) \approx 0.07\text{--}0.10$
- SNR ≥ 10 , epoch ≥ 60 s: $\text{RMSE}(\chi) \approx 0.04\text{--}0.06$
- Primary error source: Movement artifacts during rest epochs

Gait autocorrelation:

- ≥ 100 strides, $\text{CV}(\text{speed}) < 0.15$: $\text{RMSE}(\chi) \approx 0.08\text{--}0.12$
- ≥ 200 strides, $\text{CV}(\text{speed}) < 0.10$: $\text{RMSE}(\chi) \approx 0.05\text{--}0.08$
- Primary error source: Non-exponential decay in short walks

3.3.3 Clinical Discriminability

The clinical thresholds defining the adaptive window boundaries (0.8, 1.0) and pathological attractors (0.8, 1.2) are separated by 0.2–0.4 units. Given typical RMSE of 0.05–0.10, these thresholds exceed measurement uncertainty by $\geq 3\times$, ensuring:

- Reliable classification of adaptive vs. underdamped states
- Reliable classification of adaptive vs. overdamped states
- Adequate power for detecting longitudinal χ drift

For borderline cases (χ within 0.1 of threshold), confidence intervals should be computed and reported alongside point estimates.

3.3.4 Recommended Reporting

All clinical applications should report:

1. Point estimate: $\hat{\chi}$
2. 95% confidence interval: $[\hat{\chi} - 1.96 \cdot \text{SE}, \hat{\chi} + 1.96 \cdot \text{SE}]$
3. Quality metrics: SNR, epoch duration, number of cycles/strides
4. Modality: LFP, accelerometry, gait, or composite

3.4 S3.4 Information Efficiency

The PCC stages can be understood through the information efficiency functional:

$$\eta(\chi) = \frac{I(\chi)}{\Sigma(\chi)} \quad (4)$$

where $I(\chi)$ is mutual information (between input and output) and $\Sigma(\chi)$ is entropy production. Prior SymC derivations show $\eta(\chi)$ is maximal when $\chi \approx 1$. Each PCC stage represents a reduction in η :

- Stages 1–2: η decreases due to noise and variance increase
- Stage 3: η decreases due to substrate degradation (increased Σ)
- Stages 4–5: η collapses as the system loses adaptive capacity

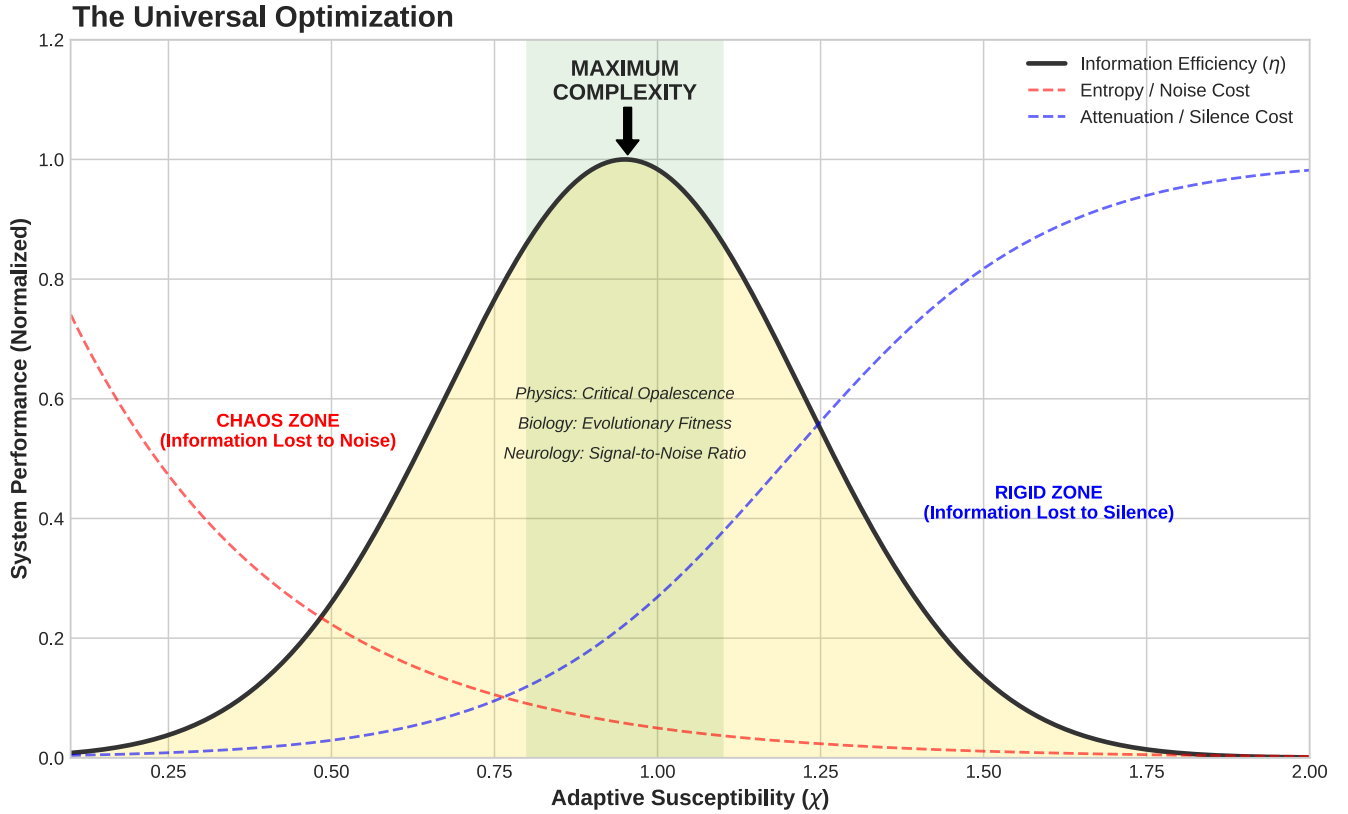


Figure 3: **The Universal Optimization Principle.** Information efficiency η (black curve) as a function of the adaptive susceptibility ratio χ . The peak at $\chi \approx 1$ represents maximum complexity, where systems achieve optimal signal-to-noise ratio. The green-shaded adaptive window ($0.8 \leq \chi \leq 1.0$) encompasses the high-efficiency plateau. Left of the window (Chaos Zone, red): entropy/noise cost (dashed red) dominates, and information is lost to noise. Right of the window (Rigid Zone, blue): attenuation/silence cost (dashed blue) dominates, and information is lost to silence. Domain-specific interpretations: Physics (critical opalescence), Biology (evolutionary fitness), Neurology (signal-to-noise ratio). The yellow-shaded area under the efficiency curve represents integrated system performance.

4 S4: Cross-Domain Mapping

The PCC framework applies across biological, geological, and engineered systems. Table 3 provides the domain-specific interpretation of each stage.

Table 3: PCC stage mapping across domains

Stage	Parkinson's	Seismology	Oncology	Viral
1. Gain Failure	Hyposmia, constipation	Microseismic rate change	Immune surveillance drop	Receptor downregulation
2. Damping Failure	RBD, sleep disruption	Viscosity shifts, creep changes	Checkpoint evasion	Innate response fatigue
3. Material Fatigue	α -synuclein accumulation	Stress accumulation, fault weakening	Tumor microenvironment remodeling	Tissue reservoir establishment
4. Resonance Nucleation	Beta bursting (13–30 Hz)	Tremor clustering, foreshocks	Metastatic seeding	Viral blips, transient viremia
5. Resonance Lock	Clinical motor PD	Mainshock rupture	Treatment-resistant progression	Rebound viremia

4.1 S4.1 Physical Interpretation by Domain

Neurodegenerative (PD, AD, DLB):

- γ = inhibitory tone (GABAergic, dopaminergic)
- ω = neural oscillation frequency (beta, alpha, theta bands)
- Failure direction: Typically χ drops (underdamped tremor) then rises (overdamped rigidity)

Seismology:

- γ = fault zone viscosity, frictional dissipation
- ω = derived from Rate-and-State Friction (RSF) Jacobian eigenvalues
- Failure direction: χ drops below 1 (underdamped “ringing”) then $\chi \rightarrow \infty$ at rupture

Oncology:

- γ = immune clearance rate, therapy-induced decay
- ω = tumor growth oscillation frequency (treatment cycles)
- Failure direction: χ drops below 0.8 predicts oscillatory relapse

Viral Control:

- γ = viral clearance rate (immune + antiviral)
- ω = viral replication oscillation frequency
- Failure direction: Inadequate γ yields $\chi < 0.8$ and rebound

5 S5: Predictive Algorithm

5.1 S5.1 General PCC Detection Protocol

1. **Baseline establishment:** Measure $\chi(t)$ during confirmed stable period. Compute mean $\bar{\chi}$ and variance σ_{χ}^2 .
2. **Continuous monitoring:** Track $\chi(t)$ using domain-appropriate instrumentation (EEG, seismometer, ctDNA, viral load).

3. **Stage 1 alert:** Signal-to-noise degradation without χ shift. Sensitivity-specific; typically subclinical.
4. **Stage 2 alert:** Variance σ_χ^2 exceeds baseline by threshold factor (typically $> 2\times$) while mean $\bar{\chi}$ remains in adaptive window.
5. **Stage 3 alert:** Mean $\bar{\chi}$ shows systematic drift from baseline (linear regression slope $\neq 0$ at $p < 0.05$).
6. **Stage 4 alert:** $\chi(t)$ exits adaptive window ($\chi < 0.8$ or $\chi > 1.2$) for sustained periods (domain-specific duration threshold).
7. **Stage 5 confirmation:** System captured in pathological attractor. Clinical/operational diagnosis.

5.2 S5.2 Preclinical Divergence Signatures

The critical slowing phenomenon provides advance warning before Stage 4:

- **Increased variance:** $\text{Var}[\chi(t)]$ rises before mean shifts
- **Increased autocorrelation:** Recovery from perturbations slows
- **Increased kurtosis:** Extreme excursions become more frequent
- **Spectral broadening:** Power spectrum shows widening around characteristic frequencies

These signatures are mathematically equivalent across domains, enabling unified monitoring algorithms.

6 S6: Falsification Criteria

The PCC framework is falsified if any of the following are observed:

1. **Stage skipping:** Systems transition directly from Stage 1 to Stage 5 without intermediate stages.
2. **Variance stability:** Systems fail without preceding variance increase in $\chi(t)$.
3. **Non-monotonic cascade:** Systems regularly reverse from Stage 4 to Stage 2 without intervention.
4. **χ -independent failure:** Failure occurs while χ remains stable within the adaptive window.
5. **Cross-domain inconsistency:** The stage sequence differs fundamentally between domains (e.g., resonance nucleation precedes damping failure).

7 S7: Relationship to SymC

The PCC is a *failure-trajectory* specialization of the general SymC boundary principle. While SymC describes the mathematical boundary $\chi = 1$ and the adaptive window $0.8 \leq \chi \leq 1.0$, PCC describes the *sequence* by which systems depart from this window en route to failure.

SymC provides:

- The critical boundary definition ($\chi = 1$)
- The adaptive window (0.8–1.0)
- The information-efficiency justification
- The substrate inheritance mechanism

PCC adds:

- The five-stage failure ontology
- Stage-specific detection signatures
- The intervention window identification
- Cross-domain cascade mapping

Together, SymC + PCC form a complete Preclinical Divergence detection architecture applicable to any adaptive system operating near the critical-damping boundary.

8 S8: Glossary

Adaptive Window The range $0.8 \leq \chi \leq 1.0$ where systems maintain optimal stability-responsiveness trade-off.

Critical Damping Boundary The mathematical boundary $\chi = 1$ separating oscillatory ($\chi < 1$) from monotonic ($\chi > 1$) dynamics.

Critical Slowing The increase in variance, autocorrelation, and recovery time that precedes critical transitions.

Damping Ratio (χ) The dimensionless ratio $\gamma/(2|\omega|)$ governing system dynamics.

Gain Failure Stage 1: Loss of input sensitivity without controller malfunction.

Damping Failure Stage 2: Degradation of inhibitory/dissipative feedback.

Material Fatigue Stage 3: Accumulation of structural/substrate damage.

Resonance Nucleation Stage 4: Emergence of pathological oscillations or critical slowing.

Resonance Lock Stage 5: Irreversible capture in pathological attractor.

Intervention Window The period between Stages 2 and 3 where cascade reversal remains achievable.

Overdamped $\chi > 1$: System responds monotonically, potentially rigidly.

Underdamped $\chi < 1$: System responds oscillatorily, potentially unstably.

PCC (Prodromal Critical Cascade) The five-stage failure sequence preceding critical system failure.

Preclinical Divergence The detectable period (Stages 2–3) when χ variance increases and systematic drift begins, prior to clinical manifestation. The target window for intervention.

SymC (Symmetrical Convergence) The framework identifying $\chi = 1$ as a universal boundary principle.

Supplementary References

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